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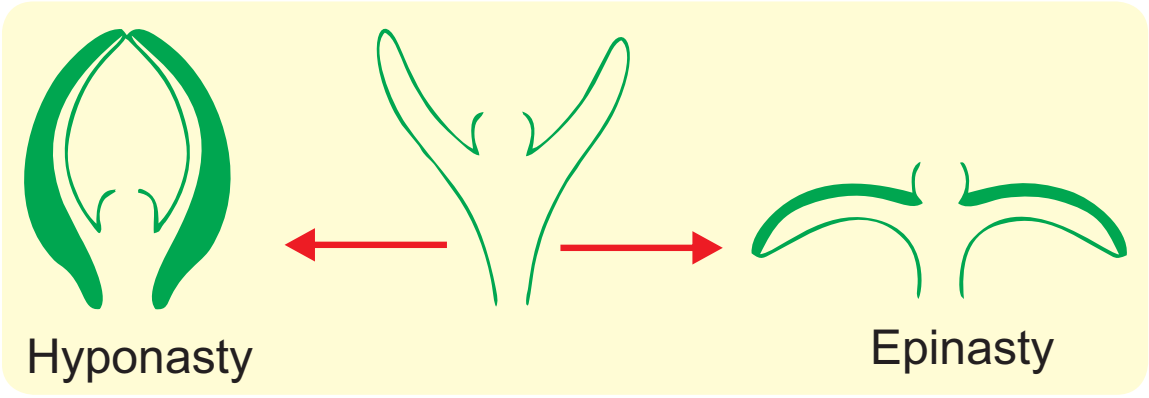


Figure 15.16: Hyponasty and Epinasty

direction. This movement is called **circumnutation**. It is commonly found in the stems of climbers of *Cucurbitaceae* (Figure 15. 17).

b. **Autonomic Movement of variation:** It happens in Indian telegraph plant. (*Desmodium gyrans*). The compound leaf consists of a larger terminal and two smaller lateral leaflets. During day time, the two lateral leaflets move upward at an angle of 90° and come to lie parallel to the rachis. Again, they may move downward at 180° so that they are parallel to the rachis. They may again move upward at 90° to come in their original position. All these movements occur with jerks after intervals, each movement being completed in about 2 minutes (Figure 15.18).



Figure 15.17: Circumnutation

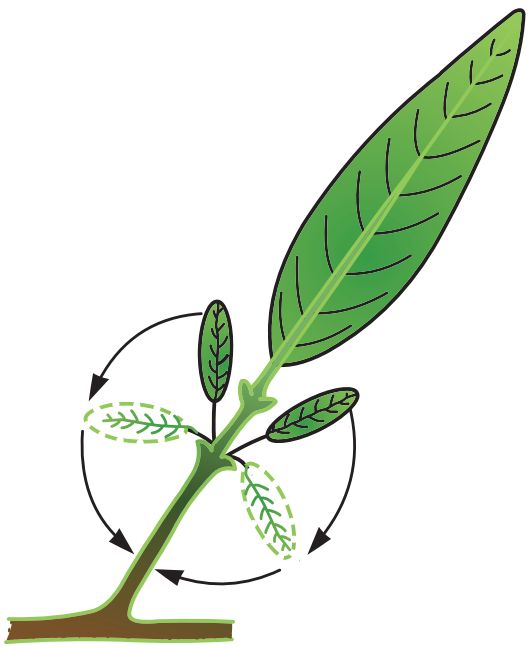


Figure 15.18: Autonomic Movement of Variation

ii. Paratonic (induces) movements of curvature

The movement arising from external stimulus is called **Paratonic** (Induced) movements of curvature. They are of two types. 1) Tropic movements 2) Nastic movements (Table 2)

a. Tropic movements

A movement that occurs in response to an unidirectional stimulus is called **tropic**